Chuanshuai (Adam) Fu

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Google Scholar | ResearchGate | ORCID

BASIC INFORMATION

Date of Birth: December 21, 1998 | Place of Birth: Hainan Island, China
Nationality: Chinese | Current residence: Edmonton, Canada | Hobby: Table tennis

EDUCATION & EXPERIENCE

- **University of Alberta (PhD in Physical Oceanography, Canada)** 09/2022-06/2026

Thesis: Exploring the dynamics and thermodynamics of the Arctic Ocean's shelf seas using ocean model simulations and satellite altimetry

- Alfred Wegener Institute (Research intern, Germany) 02/2026-04/2026
- Alfred Wegener Institute (Research intern, Germany) 05/2025-07/2025
- Ocean University of China (Visiting student, Qingdao) 12/2024

- **University of Alberta (MSc in Physical Oceanography, Canada)** 09/2020-07/2022

Thesis: Modelling the Atlantic Water along its poleward pathway into and through the Arctic Ocean

- **East China Normal University (BSc in Physics, Shanghai)** 09/2016-07/2020

Thesis: Sensitivity of Baffin Bay to exchanges through its gateway straits

- National Chiao Tung University (Exchange student, Taiwan) 09/2018-01/2019
- University of Alberta (Research intern, Canada) 07/2019-09/2019

RESEARCH INTERESTS

Physical oceanography, Ocean modelling (e.g., NEMO), Satellite altimetry (e.g., SWOT), High-latitude ocean circulation, Eddy-sea ice, air-sea, and glacier-ocean interactions, Arctic heat and freshwater budgets, Arctic and Atlantic Meridional Overturning Circulation

PUBLICATIONS

The changing seasonality of mixed layer temperature in the Arctic shelf seas in response to surface heat flux forcing (**Chuanshuai Fu***, Wenli Zhong, Clark Pennelly, Paul Myers, under review in *Ocean Modelling*)

Fu, C.*, Pennelly, C., & Myers, P. G. (2026). Does the Beaufort Gyre freshwater storage directly respond to changes in Bering Strait inflow? Insights from NEMO modeling experiments. *Journal of Geophysical Research: Oceans*, 131, e2025JC023782. https://doi.org/10.1029/2025JC023782

Fu, C., Han, X., Wang, Q.*, Pennelly, C., & Myers, P. G (2026). Wide-swath satellite altimetry reveals hotspots of small mesoscale eddies in the western Arctic Ocean. *Communications Earth & Environment*. https://doi.org/10.1038/s43247-026-03498-9

Fu, C.*, Müller V., & Myers, P. G. (2025). Large mesoscale eddy properties and dynamics in the Labrador Sea from satellite altimetry, *Atmosphere-Ocean*, 1-19. https://doi.org/10.1080/07055900.2025.2538892

Fu, C.* & Myers, P. G. (2024). Exceptional sea ice loss leading to anomalously deep winter convection north of Svalbard in 2018. *Climate Dynamics,* 62, 2349–2367. https://doi.org/10.1007/s00382-023-07027-8

Fu, C.*, Pennelly, C., Garcia-Quintana, Y., & Myers, P. G. (2023). Pulses of Cold Atlantic Water in the Arctic Ocean from an ocean model simulation. *Journal of Geophysical Research: Oceans*, 128, e2023JC019663. https://doi.org/10.1029/2023JC019663

Myers, P. G.*, Castro de la Guardia, L., **Fu, C.**, Gillard, L. C., Grivault, N., Hu, X., et al. (2021). Extreme High Greenland Blocking Index Leads to the Reversal of Davis and Nares Strait Net Transport Towards the Arctic Ocean. *Geophysical Research Letters*, 48, e2021GL094178. https://doi.org/10.1029/2021GL094178

POSTERS

Hotspots of small mesoscale eddies in the western Arctic Ocean observed by SWOT (Ocean Sciences Meeting & Polar Heat Workshop, Glasgow, UK; 02/2026)

Insights from NEMO simulations on the recent hydrographic changes in the Beaufort region and the Bering Strait (23rd Arctic-Subarctic Ocean Fluxes workshop, Barcelona, Spain; 05/2025)

Insights from NEMO simulations on the recent hydrographic changes in the Beaufort region (ICTP-SORP-NORP Summer School and Workshop on Polar Climates, Trieste, Italy; 07/2024)

The wide warming of the shallow shelf seas in the Arctic Ocean during 2014-2019 versus 2008-2013 (CAMAS workshop, Santa Fe, United States; 02/2024)

Eddy properties and dynamics in the Labrador Sea from satellite altimetry (Ocean Sciences Meeting, New Orleans, United States; 02/2024)

Exploring the Sources and Fate of the Cold Atlantic Water (CAW) from Ariane experiments (ArcTrain Annual Meeting, Montreal, 05/2022)

Modelling the Atlantic Water Inflow at the Gateways of the Arctic Ocean with Two Different Simulations (ArcTrain Annual Meeting, Isle-aux-Coudres, 10/2021)

Modelling Heat and Salt Fluxes across Davis Strait, 2004-2016, using the ANHA12 Configuration (University of Alberta internship poster symposium, 08/2019)

PRESENTATIONS

Wide-swath satellite altimetry reveals hotspots of small mesoscale eddies in the western Arctic Ocean (24th Arctic-Subarctic Ocean Fluxes workshop, Bergen, Norway; 04/2026)

Hotspots of small mesoscale eddies in the western Arctic Ocean observed by SWOT (section meeting, Bremerhaven, Germany; 07/2025)

Insights from NEMO simulations on the recent hydrographic changes in the Beaufort region and the Bering Strait (Drakkar workshop, Grenoble, France; 02/2025)

The changing seasonality of mixed layer temperature and its driving mechanism in the Arctic shelf seas (IGR Fall Symposium, University of Alberta; 11/2024)

Eddy properties and dynamics in the Labrador Sea from satellite altimetry (IGR Winter Symposium, University of Alberta; 04/2024)

Exceptional sea ice loss leading to winter deep convection north of Svalbard in 2018 (North Research Day, University of Alberta; 03/2023)

Exploring the dynamic and thermodynamic processes causing the winter deep convection north of Svalbard in 2018 (Best Oral Presentation [First Prize]; IGR Fall Symposium, University of Alberta; 11/2022)

The Variability of Two Atlantic Water Branches through Fram Strait and the Barents Sea (Canadian Meteorological and Oceanographic Society; Online; 06/2022)

Modelling the Atlantic Water along its Poleward Pathway into and through the Arctic Ocean (Drakkar Workshop; Online; 02/2022)

AWARDS & TRAVEL FUND

Chinese Scholarship Council (CSC) Scholarship (09/2022-08/2026)

Alberta Graduate Excellence Scholarship (11/2025)

Steve and Elaine Antoniuk Graduate Scholarship in Arctic Research in Earth and Atmospheric Sciences (09/2025-08/2026)

Mitacs Globalink Research Award (11/2025)

Education Abroad Individual Award (03/2025)

Indira V Samarasekera Scholarship in Global Education (03/2025)

Patricia Anne Cavell Graduate Scholarship in Earth Sciences (11/2023)

Roy Dean Hibbs Memorial Scholarship (05/2022, 04/2023, 05/2025 [with distinction])

Roy Dean Hibbs Memorial Travel Award (11/2021, 05/2022, 11/2023, 04/2024, 04/2025, 11/2025)

American CAMAS Travel Support for Early-career student (02/2024)

Italian IASC Travel Fund for Young Scholar (07/2024)

Canadian Ocean Modelling Forum Workshop Travel Funding (12/2025)

Norwegian Travel Support for Early Career Researcher (02/2026)

ADDITIONAL INFORMATION

Computer skills: MATLAB, Python (e.g., Jupyter notebook), HPC (The Shell and Scheduler)

Languages: Mandarin & Hainan dialect (native), English (proficient), Cantonese (basic)

Journal reviewer for: Nature Climate Change (two manuscripts)

Teaching training: Graduate Teaching and Learning Program Level 1: Foundations

Teaching assistant for: Planet Earth, Climate system, Atmospheric modelling, The ocean, Violent weather

Courses: Reading and seminar courses, Advanced atmosphere-ocean dynamics, Atmospheric modelling, Fluid dynamics, [Self-directed learning: Descriptive Physical Oceanography, Data analysis in ocean sciences, AI and machine learning in marine science]